

Format for Project Proposal under Technology Development in FinTech

1. Title of the Project:

“Development of Secured Voice based AI-enabled Digital Payment Application for Indian Regional Languages”

2. Thematic Area Under Which Your Project Lies? (IoT/AI/ML/Blockchain/Payments System/ (If Others, please specify): AI/ML/Payments System

3. Proposed Duration of the Project: 24 Months

4. Principal Investigator (PI) and Co-Principal Investigators (Co-PIs)

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5. Aims and Objectives

1. To develop a user friendly secure application for digital payments in regional languages using Artificial Intelligence for the rural population to empower the digital villages.
2. To build a complete voice-based payment model with multilingual support using Natural Language Processing to perform banking-related activities for disabled and blind people in their regional language.
3. To develop a robust regional language support-based audio warning system that pops up during concurrent call and transaction processes and alerts the user of debit transactions.
4. Development of a visual/GIF platform integrated with an audio warning system to replace textual transaction detail for better clarity to the user about debit transactions.
5. Development of alternative multi-tiered authentication schemes to OTP for secure online internet banking transactions such as QR code, device/location-based authentication, etc.

6. Context/Background [Please elaborate on the existing literature relevant to the project?]

ANZ was the first Australian Bank to present voice biometrics for mobile banking in September 2017. The software picks up very instant differences in the shape of a person's vocal cords or how their nasal passages work. The audio file is completely unlike when a person's voice is recorded and played. The algorithms within the software pick up the differences in that audio file and can recognize that, as per Robert Schwarz [1].

HSBC's banking app allows customers to sign in by writing their birthdate and account specifications and saying, "My voice is my password" on the mobile device. The company says it computes 100 diverse attributes of the individual's voice to approve a user's identity [2]. According to a recent HSBC UK analysis, voice recognition technology prevented over 50% of banking scams in 2020, protecting clients' bank savings worth £249 million [3].

The three best-known voice assistants, Alexa, Siri, and Google Assistant, have been incorporated into services by the Australian Westpac Banking Corporation [4] for customers to access their accounts. The internet-based electronic payment technique can be conducted by implementing a two-system method based on an artificial neural network model (ANN): coding and validation systems. Text data (TD) comprises information like names, dates,

amounts, etc. and speech data (SD) relates to voiceprint-recorded signals [5]. For example, Venmo, Square Cash, and PayPal offer voice payments to their customers.

Voice is considered an essential biometric, and multimodal recognition systems are springing up to improve the robustness of authenticity [6]. Speech recognition can be extended to identify speakers, exploiting the information present in the speech, and various methods, including exploiting the excitation source [7, 8]. In 2011 Apple launched Siri, similar to Google's Voice Search. And In 2014, Amazon introduced Alexa & the Amazon Echo [9].

In India, banks are testing voice-based banking solutions. However, that is still in the infancy stage. For the rural population in India, voice-based digital banking solutions may play a crucial role in realizing digital India in smart villages, as villagers may not be tech-savvy. However, security against electronic frauds (e-frauds) is a major challenge in realizing digital banking in India. According to National Crime Records Bureau (NCRB), India reported 50,035 cybercrimes in 2020, 44,546 cases in 2019 and 27,248 cases in 2018. The year 2020 saw 4,047 cases of online banking fraud, 2,160 cases of ATM fraud, 1,194 credit/debit card fraud and 1,093 OTP frauds. There were also 972 cases of cyber stalking/bullying of women and children and 578 cases of fake news on social media, NCRB data showed. It can be inferred from the above data that OTP frauds constitute 25% of banking frauds. So, it is the need of the hour to make the OTP-based system robust enough to prevent such crimes. We have found a few research papers suggesting ways to avoid OTP fraud.

From the research perspective also, there are minimal works in this area. To avoid online transaction fraud happening due to static usernames and passwords [10-11], a dynamic username-based approach was proposed by the authors in [12] that utilizes different cryptographic primitives, such as encryption, digital signature, and hashing. Nowadays, a few banks use two-factor authentication before initiating a financial transaction. OTP is sent on mobile phones and needs to be entered for successful login into your bank account. However, in such cases, also fraudsters get the excess of OTP by impersonation/ misleading unaware users. In another way, if any fraudster gains access to the user's screen (though screen share), then OTP can be easily assessed, and fraud cannot be prevented.

To prevent OTP fraud, shubhangi et al. [11] have proposed an extended visual cryptography (EVC) or QR code-based method for OTP generation and verification. One of the recent advancements in the e-fraud prevention and management area, one of the leading IT industry Infosys, presented a white paper in 2018 [12], which provides key recommendations for financial institutions (FIs) to transform their antifraud technology solutions effectively. One of the key recommendations is using a multi-tiered approach in place of the existing single-tier approach to prevent e-frauds.

Taking clues from the above literature survey, we propose a secured voice-based AI-enabled digital payment banking solution in this proposal to minimize and prevent the risk of ever-increasing e-frauds. We aim to develop a robust local language support-based audio warning system to guide and alert the user during debit transactions. We aim to secure online internet banking/browser-based transactions by linking the mobile banking app to a particular online

banking session by validating through different means such as QR-code, device and location authentication, etc.

7. Problems to be addressed [Technical fallouts in the existing state-of-the-art methods that are proposed to be addressed]

EXISTING STATE OF ART

The following are currently existing state-of-the-art methods for E-payment:

7.1. Unified Payments Interface (UPI):

The Unified Payments Interface (UPI) is a system that integrates various bank accounts, seamless fund routing, and merchant payments into a single mobile application (of any participating bank). Some popular UPI apps are PhonePe, Paytm, Google Pay, BHIM, Mobi Kwik, Uber, SBI Pay and BOB UPI. A universal interface created and hosted by NPCI is the foundation for UPI. This common layer manages transactions using IMPS and the Aadhaar Enabled Payment System and ensures settlement across bank accounts (AEPS). To facilitate transactions through Virtual Payment Addresses and prevent the need to communicate account information or login credentials, banks, financial institutions, and other businesses offer UPI services linked to the NPCI's unified interface through standard APIs. The UPI system always uses a personal phone for payment authentication and authorization. Any-to-any interoperable payments can be made using a common set of APIs since this layer provides a single interface [13]. The basis of UPI is the idea of a virtual payment address. A specific virtual payment address can be mapped to bank accounts, cards, and wallets. An account number, mobile number, and Aadhaar number can all be used to make payments (virtual payment address). UPI uses the infrastructure that already exists for authentication.

7.2. Net Banking:

Internet banking is another name for net banking. Banks offer their clients an electronic system that enables them to access financial and non-financial banking services quickly. The net banking system does away with the necessity to go to the bank in person. Customers can conveniently access financial services through net banking, including money transfers, the construction of FDs and RDs, the tracking of transactions, etc.

Internet banking customers may access the bank's Web site using any device that can connect to the Internet through the access network. At the Web service provider's network, a Web server hosts the Web site of the bank. The customer accesses the Web site using one of the Web browsers. The Web service provider's network is connected with the bank's network through a secured, point-to-point connection. A third party may provide this connection. Through this secured, private connection, the Internet banking server (which resides between the Web server and the bank's internal network) transfers sensitive information that performs transactions, account balance, bill payment, etc. Hence, the customer's account-related information is never transferred through the public Internet, and the customer's account's security is maintained [14].

7.3. PoS Terminals:

A point-of-sale (POS) terminal is used in retail establishments to handle card payments. Hardware includes inbuilt software for reading magnetic strips on credit and debit cards. However, due to digitization, the reach of PoS is growing, and this service is now accessible via internet browsers and mobile platforms. PoS terminals come in various forms, including physical, mobile, and virtual. It works with multiple terminals connected to a file server located throughout the store. Every terminal in the shop receives the essential configurations set on the server. POS systems are integrated with third-party payment providers that accept various payment methods since they accept a variety of payment methods themselves.

After reviewing the existing state-of-the-art solutions, we identified the following limitations/loopholes that need to be addressed:

- None of the existing payment methods has regional language support.
- Physically disabled and blind people may not be able to operate these devices properly.
- Even if these solutions support multiple languages, illiterate and rural people often need special guidance to use them as they cannot understand the UI of the device since they are not used to such technology.
- Also, it would be easier and more convenient for the end user if he could operate the devices and make payments using voice commands rather than doing everything manually.
- Most e-fraud transactions happen during a call where a fraudster disguises himself as a bank person and asks for the user's bank account details, followed by OTP to carry out specific banking-related tasks. The reason for such frauds is the misguided directions and threat (of account closure or card blocking, etc.) received by the fraudulent caller for doing a fraudulent transaction.
- The second most common form of fraudulent transactions is link/QR code-based transactions in UPI transactions using mobile banking. Here, the user gets the payment link/QR code, and the fraudulent caller lures them to receive money in the form of some prize, lottery, etc., through clicking the link/QR code and entering the transaction/UPI password. The user often enters their passwords out of greed to receive money; instead of receiving money, they lose money. Although the transaction info, in small font size text, is always there in the payment apps, the app users either overlook it in haste or get manipulated by the fraudster to do a debit transaction by entering the UPI pin in the false appearance of checking the account balance/receiving prize/lottery etc. by entering the UPI pin.
- There is no robust authentication mechanism for browser-based online transactions except OTP-based authentication exercised by a few selected banks. However, in

case of a compromised mobile screen or cloned SIM, the OTP-based authentication may fail.

8. The novelty of the Project [Mention how the proposal is novel with respect to the current state-of-the-art FinTech?]

1. We propose a complete voice-based secured payment model with multilingual support for different regions of India, using Artificial Intelligence and Natural Language Processing.
2. Primarily rural residents of India would be able to make payments in their respective local languages and know their bank balances. They can check their transaction history and make transactions by speaking (voice input) in their regional language.
3. We use a Two-factor authentication approach to provide the extra functionality, which includes fingerprints and facial recognition in the application. Also, recorded voices can be recognized by detecting live speech.
4. Since this system is entirely voice-based, everyone, including illiterate, blind and physically-challenged people, would be able to use it without facing any difficulties.
5. Development of a robust local language support-based audio warning system integrated with a visual/GIF platform that pops up during concurrent calls and transactions. In the case of the standalone transaction, it will also guide the transaction process and alert the user about debit transactions.
6. Development of alternative multi-tier authentication schemes based on QR-code, device/location, etc., for secure online internet banking transactions.

9. Methodology

We may use Amazon Alexa's functionality to create a speech-based payment service in regional languages since it also supports many regional Indian languages. Alternatively, we can use the ASR (Automatic Speech Recognition) paradigm to create a voice assistant. Initially, the user calls an assistant and sets a task. The assistant then transmits the message to NL tools, where filtered intent and entities are sent to the bank-developed and hosted Bot framework/database. Building a virtual voice assistant is the most crucial step in creating a voice-based payment solution for localised languages. We need to collect data on Indian voices in the local tongue and train the model to construct a voice assistant.

9.1. System design using existing approaches:

The functioning 'Amazon Alexa' can be used as it also supports many Indian regional languages. The following job is "Wake Word Detection", which defines whether the user says one of the words the device is programmed to turn on. This is complicated as it requires identifying pronunciation differences on a limited processing power device. If the

wake word is identified, the signal is then conveyed to the speech recognition software in the cloud, which takes the audio and transforms it into a text format. In the beginning, the system accepts natural language as an input.

9.2 System design using Automatic Speech Recognition (ASR) approach:

ASR uses computers to convert spoken words into written ones and is the main principle behind AI-based voice assistants [15]. The working of ASR is shown in figure 1. Firstly, Analog to Digital Conversion is done where in most cases, speech is typically recorded and made available in analogue format. Standard sampling methods or tools can be used to digitize analog voices using sample and quantization techniques.

Digital speech is usually represented as a one-dimensional vector of voice samples, each of which is an integer. Speech pre-processing includes detecting and eliminating silent frames and methods for reducing or removing noise through signal processing. The speech is split into 20-second frames after pre-processing to facilitate feature extraction later. Then Feature Extraction is done where speech frames are converted into a feature vector that specifies which phoneme or syllable is being spoken [16].

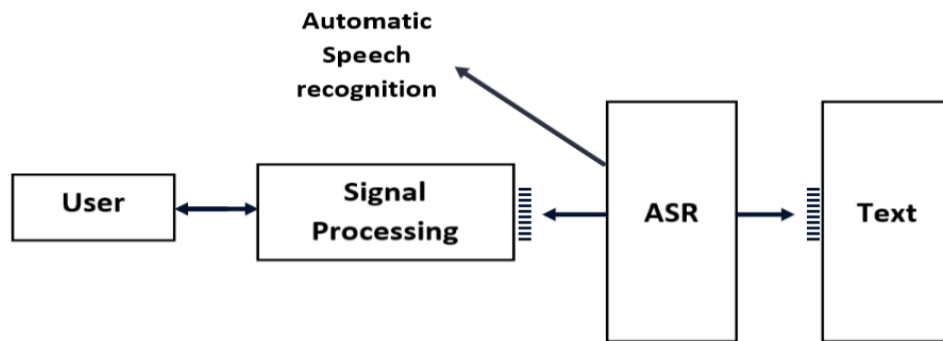


Fig. 1. Working process of ASR

There are many speech recognition software, but all the work happens in its ASR. In a nutshell, the system begins with the device gathering audio from the source, where the source is a microphone. Recorded speech waveforms will be sent to acoustic analysis, performed on three levels, as shown in figure 2.

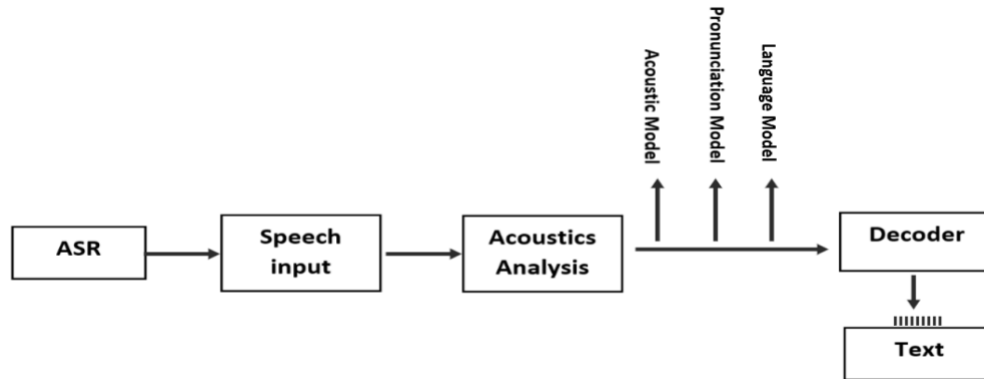


Fig. 2. ASR work-flow

9.2.1 The three modelling are as follows:

- **Acoustic Modelling:** It represents whether the elements were pronounced and the words that can complete them.
- **Pronunciation Modelling:** Analyses of how these elements are pronounced will check whether there are accents or other peculiarities.
- **Language Modelling:** It is usually trained toward finding contextual probabilities by counting on what elements were gained. All the recorded data gets processed by Artificial Intelligence without any human interaction. The speech waveform data is transferred to the decoder. It finally converts into text for further use as a command.

9.2.2 The Proposed Voice Recognition System

The proposed system is characterized by pre-processing before the feature extraction, which is included in any general VRS. Pre-processing prevents the error estimation caused by speakers' volume changes. In VRS, two basic procedures are performed: feature extraction and classification. In feature extraction, catch the vital characteristic that distinguishes one speaker from the other.

Fourier Transforms (FTs) and Short-Time Fourier Transforms (TFTs) were used to extract features. However, these techniques are unsuitable for representing non-stationary signals, such as voice signals. Using (FTs), the input signal was converted from the time domain into the frequency domain to know the signal frequencies but not when. Stats tried to handle the problem by using a local window to display the frequency and time relation. However, its efficiency depended on the used window. Wavelet Transforms (WTs) has fingered some of these problems.

9.2.3 Database:

The speech signals used in this research can be drawn from the approved ELSDSR database. It is non-native speakers' database. It provides speech data for the development and evaluation of automatic VRS.

9.2.4 Pre-Processing of the Speech:

To avoid inaccurate estimation caused by speakers' loudness settings, Signals were completed by applying normalization to the mentioned speech signals before feature extraction. The architecture of the Voice Recognition System is shown in figure 3.

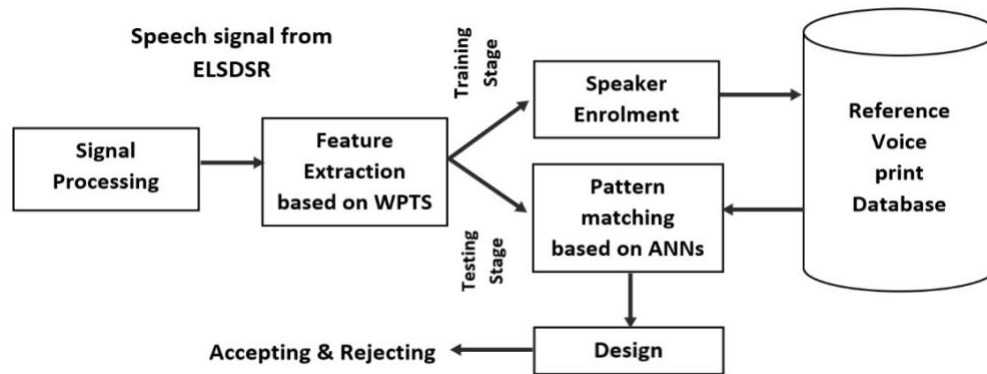


Fig. 3. Voice Recognition System

9.2.5 Feature Extraction by Wavelet Transforms:

In Feature Extraction, the signal is first transformed from the Time-domain into the Frequency Domain. Then the signal analysis is done using various filters to extract a vital feature in feature extraction. As there are multiple languages, choosing a feature that carries much data about each speaker is hard. Still, an essential feature can be extracted using STFT, which calculates the MFCCs, the familiar parameter in the speech recognition job. However, MFCCs were missed in the background noise, as specified before. The wavelet converts proved an effective technique for various signal processing difficulties. Then, the signal is decomposed using a fit wavelet transform version and calculating the energy enclosed in each signal as a differentiating feature for each speaker.

9.2.6 Classification by Artificial Neural Networks:

A universal method of communication is speech. Recognizing the speaker based on the vocal characteristics of a given utterance is known as speaker recognition (SR). On the other hand, speech recognition limits the identifying process to the content rather than the speaker. The foundation of SR is identifying and extracting distinctive features of the speaker's speech. It is a classification procedure of feature extraction using DWT and WPT and judges the extracted features' effectiveness by distinguishing between the speakers. In this research work, the classifier can be used as ANNs for the computer to learn things and make decisions in a way similar to that of a human. Artificial neural networks are attempts to replicate the network of neurons that make up the human brain. Programming standard computers make ANNs behave like networked brain cells. Here the FFBPNN can be practiced, which refers

to the multilayer networks. The error Back-Propagation training algorithm can also be used. The recognition rate can be evaluated by sketching the confusion matrix.

9.3 Design of system security protocols:

So far, we have discussed voice-based digital payment systems design methodologies. However, digital payments are prone to many security concerns and must be protected against ever-increasing e-frauds. The following objectives in this proposal are to design novel security schemes for digital payments. A flow diagram for the proposed security schemes is given in Fig.4. The proposed methodology works for both types of transactions, Mobile banking and Online/browser-based transactions, as shown in Fig. 4. On the mobile banking platform, our primary contribution lies in developing a robust audio-based warning system that guides and warns a user during debit transactions in the native language of the user. Moreover, as shown in the flow diagram, the online browser-based login/transaction is secured using a multi-tiered authentication approach, as suggested in the white paper by Infosys [12]. However, at the same time, we may leave this flexibility to the user to choose the level of authentication they desire to maintain the ease of transaction for low-risk users.

9.3.1 For Mobile Banking:

- a) To stop fraudulent call-based transactions, we want to develop a local language support-based audio warning system which will continuously monitor the payment app, phone call and text messages of the user. Suppose the user receives an OTP message and tries to submit it in the payment app, and at the same time, the user is on call. In that case, the warning system will get activated automatically (taking over the audio access of the device despite being on call/call may be disconnected) and warn the user about the possibility of the current transaction being a fraudulent transaction in their local language. However, suppose the user still wants to proceed with the transaction after the warning. In that case, the warning system continues to guide them through the transaction process about debit transactions with the amount and recipient details, as explained in figure 4.
- b) For link/QR code-based fraud transactions, we are proposing two solutions: firstly, we propose to improve the payment interface by depicting the transaction process through visuals in place of texts that may include a GIF representation of the transaction process as there is a well saying that one picture is worth a thousand words and it is a common experience that picture/GIF draws the user's attention more than a plain text message (prevailed in existing interfaces). Secondly, the moment user receives the OTP message and navigates to the payment page. The smart audio warning system will caution the user in their native language that the amount 'X' will be debited from their account after entering the OTP to person 'Y'. The above solutions are shown on the right-hand side of the flow diagram in Figure 4.

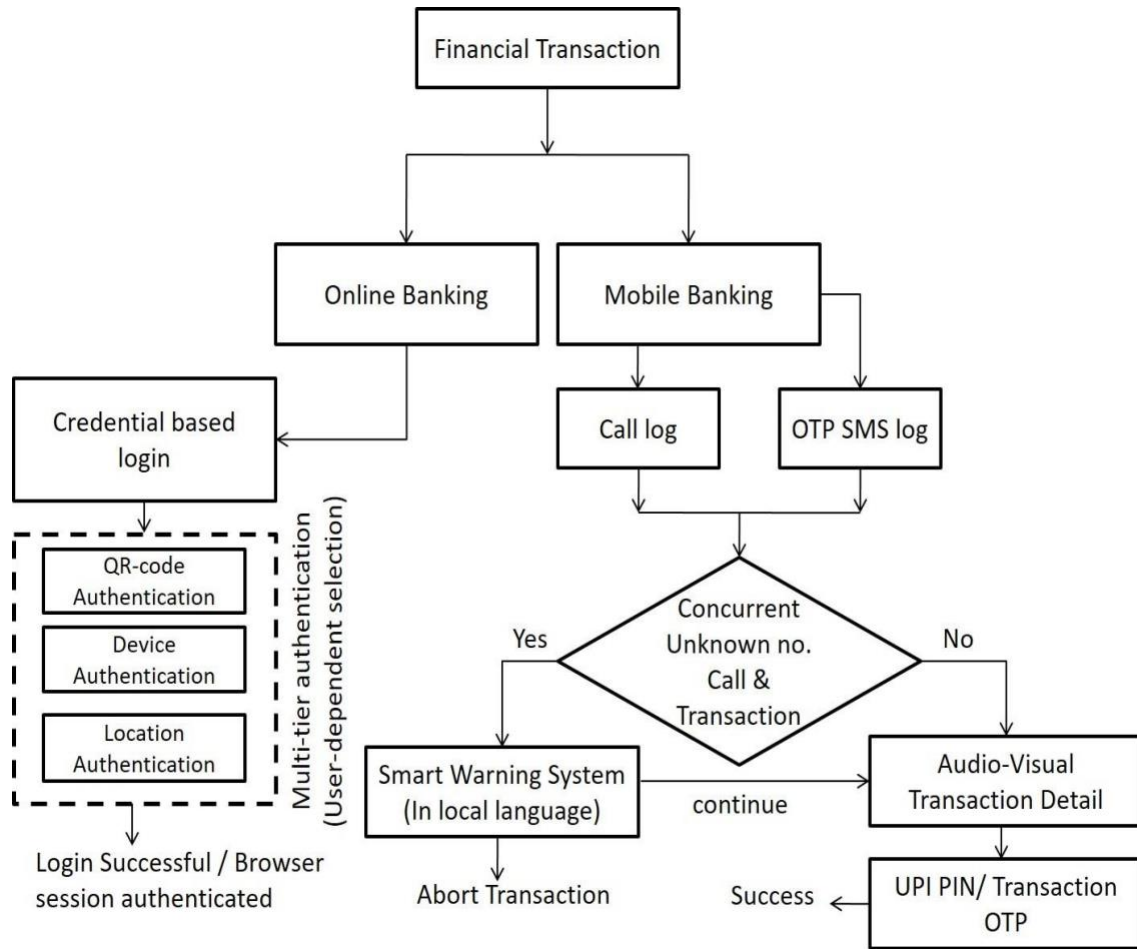


Fig. 4: Flow diagram of the proposed e-fraud prevention solution

9.3.2 For Internet Banking (Browser-based transactions):

We propose following multi-tier authentication processes to authorize a particular session for browser-based online transactions. We propose to link the mobile banking app to the user's online banking session. These measures are additional to username and password. It is noteworthy that more security features minimize the user's experience and so we may keep the following additional authentication steps optional to the user.

- As an alternative to OTP-based login authentication, we propose a QR-code-based authentication process using the mobile banking app for browser-based online transactions. Presently, a similar method is used by WhatsApp to connect your mobile app to the web-based usage of WhatsApp. We want to extend it further for payment apps.
- Further, we propose a device/location-based authentication for a browser-based transaction. If the user's online account is being logged in from an unknown device and at

a different location (than the user's current location), the payment app installed on the user's mobile device will get a notification asking the user to authenticate the browser login. The browser login would not be successful until the user verifies and allows it through their mobile payment app.

Smart Voice-based Warning System

One of the significant components of the proposed smart audio warning system is text-to-speech synthesis. The proposed warning system will read incoming text messages and extract useful financial information. One of the simplest ways could be to convert the incoming transaction-related text message into audio or speech using the standard text-to-speech synthesis procedure [17]. However, this may not be useful in the current context where the audio warning system has not just read the text message as it is but has to take a smart decision about the transaction process based on the received text message. Natural language processing (NLP)-based smart decision-making may help us to synthesize the desired audio/speech warning message based on the contents of the input text message. When an OTP-based text message is received for an amount of 'X', this will be analyzed by the proposed system, and the corresponding simplified audio message in the native language of the user will be synthesized. The system will play this message to warn the user when they reach the transaction page and try to input the OTP. NLP-based solutions may also alert the user to be cautious about any malicious links frequently received on the user's mobile as a text message (with warning contents that the user's account/card will be blocked if they don't update it by clicking the received link). They receive which are usually sent by fraudsters to steal the users' login details.

10. Background work is already done: [This section should elaborate on the business development strategy, product viability, commercialization, etc.]

Once the proposed solution is ready, it can be easily extended in a product/software form, which can be quite helpful for the existing payment banks in enhancing their e-fraud prevention and management system. The software solution can be branded and commercialized into a business form. The startups working in the area can be collaborated to extend the proposed solution further to develop more security solutions for e-fraud prevention and management of financial institutions.

10.1 Business Development Strategy

The detailed business development strategy of the proposed voice-based payment application is described in terms of proposed problem definition (Figure 5), Market Size (Figure 6), Value Proposition Canvas both for Banks and Customers (Figures 7 and 8), and Business Model Canvas (Figure 9).

Problem Definition

To develop an AI-enabled voice based mobile application for digital transactions with regional language support for illiterate, rural people and people with disabilities.

Lack of Awareness

- In the existing scenario, people manually go to the bank to get the banking service like money transactions. People get assistance through the Bheem app, Google pay, and phone pay, which is helpful for only literate people. That is not for those who do not know read and write.

Social Impact

- Especially in India, where the literacy rate is low, the significant population is unaware of using computers and the internet. Then in such a case, how can we expect such people to use the applications developed in those environments?

Alternatives Shortcomings

- Traditional digital payment is a reliable method, but it necessitates constant checking of account numbers, cross-checking among applications, among other things.
- Alternative options do not offer voice payments to their customers. They do not support Indian regional language making it inconvenient and unhandy experience.

Fig. 5. Problem Definition with Social Impact

Market Sizing

Top Down Approach

Total Available Market (TAM):-

- Market size of Digital Payments is 7.4 billion in india expected to grow at 19 compound annual growth rate to 32 billion by 2032

Serviceable Available Market (SAM):-

- Our target customer is people in regional areas of India who speaks native languages only, illiterate and physically impaired people, which comprises to 90% of the given market size comes around 6.66 billion.

Serviceable Obtainable Market (SOM):-

- We consider that 20 percent of that will consider having desire for online home health care system, which equals to around 1.33 billion

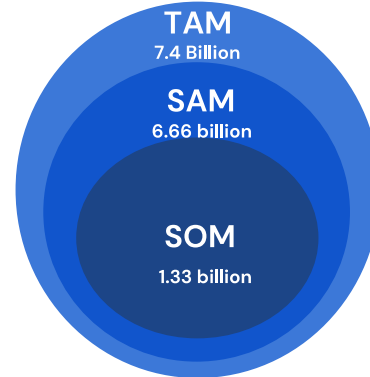


Fig. 6. Market Analysis of the Proposal

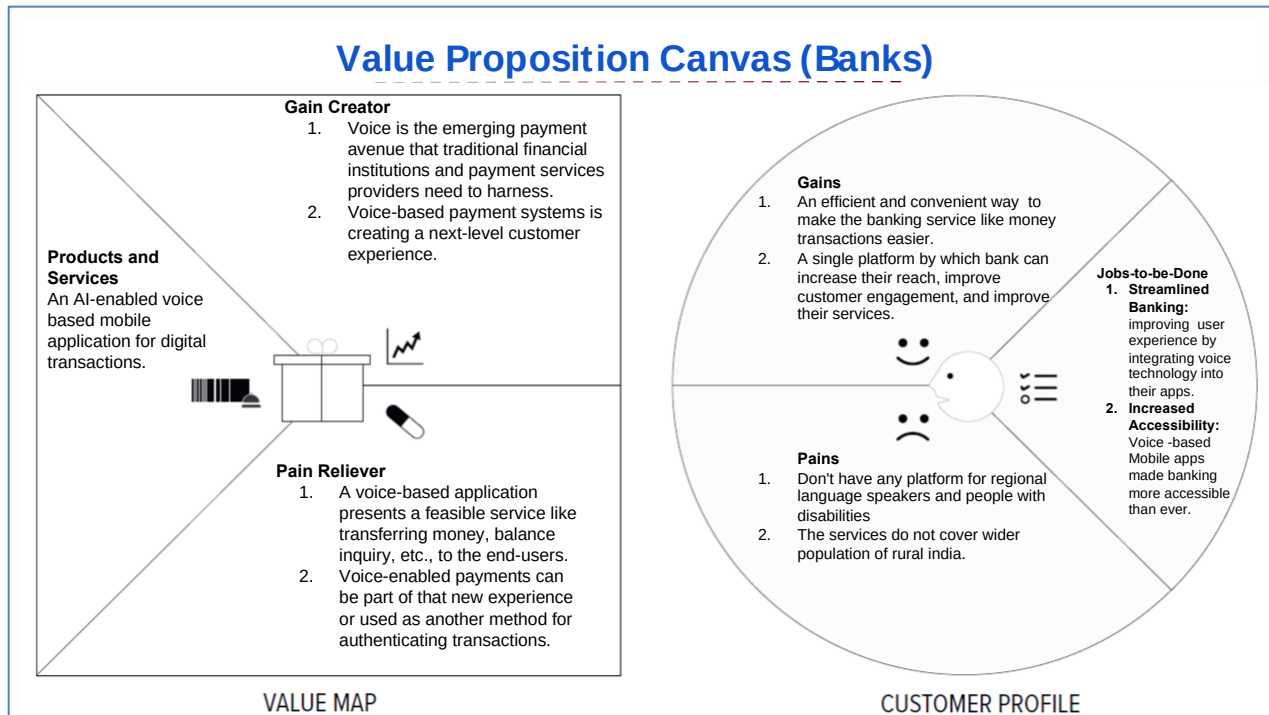


Fig. 7. Value Proposition Canvas for Banks

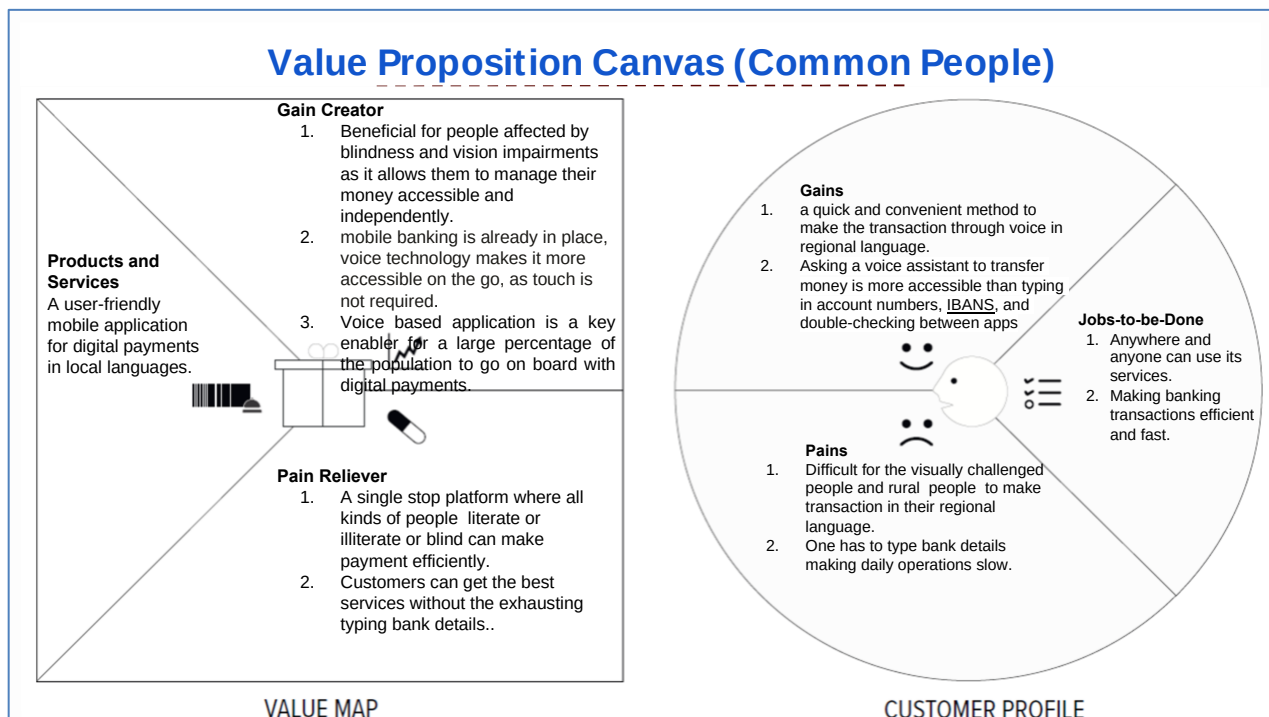


Fig. 8. Value Proposition Canvas for Customers

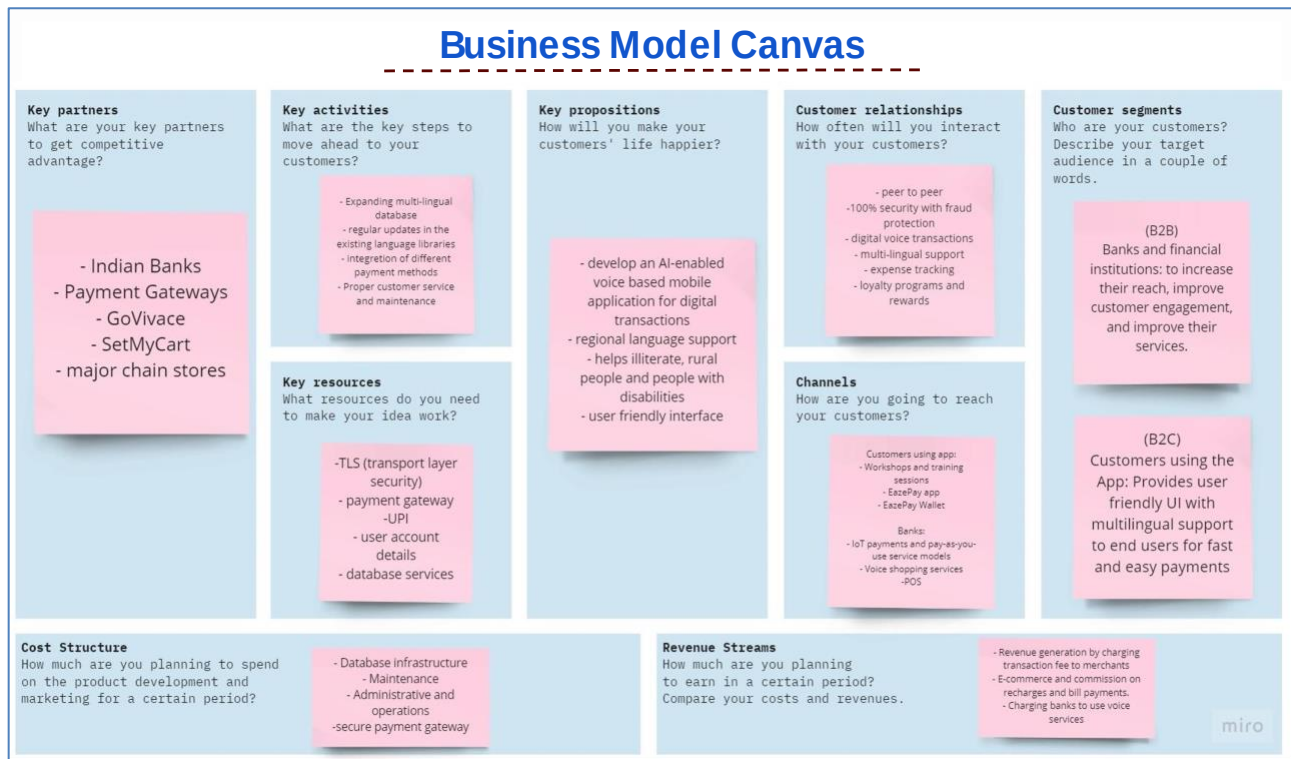


Fig. 9. Business Model Canvas of the Proposal

Revenue Sharing Mechanism:

IIIT – NR being the technology owner, Chhattisgarh Cooperative Bank and the Union Bank of India will be the knowledge partners of the proposal. GoVivace and Set-My-Cart will be the industry partners for execution and marketing. **IBITF, the sponsoring agency, will get a revenue share of 8.9% upon successfully commercializing the proposed product. The detailed revenue-sharing model and financial planning of the proposed project are given in a separate excel sheet attached as Annexure-IV.**

10.1.1 Product Development

At the core of any voice, payment is a 'skill'. A skill is similar to an app on the smart speaker's voice assistant. Using their voice, account holders can use skills to carry out banking-related actions, including checking their account balance and starting payments. Figure 10 depicts the proposed model showing how the product development will occur.

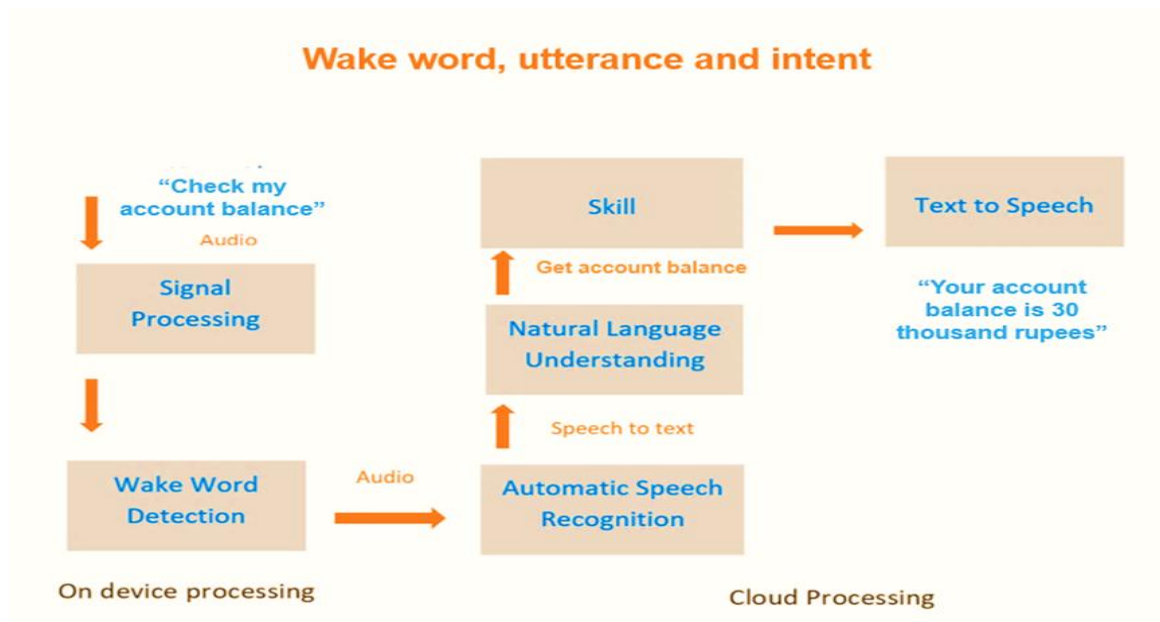


Fig. 10. Block Diagram of Proposed System

10.1.2 Industry Collaborations

1. GoVivace

- IIIT Naya Raipur is having MoU with GoVivace (<https://www.govivace.com>) for academic and industrial projects.
- It deals in speech recognition, speaker identification, voice authentication, speech synthesis and analytics, gender identification, language identification, etc.
- We will use GoVivace's speech processing-based product development expertise in implementing our project.
- We will also leverage their wide industrial reach to commercialize and monetize our product.

2. SetMyCart

- SetMyCart is a startup providing e-commerce platforms to small and medium business owners.
- SetMyCart has expressed interest in our project and ensured help in its implementation and commercialization in the e-commerce area.

3. Banks

- Union Bank of India
- Chhattisgarh Cooperative Bank

10.2 DIVERSIFICATION

There are many ways in which this application can be diversified in the future. These include P2P payments, Voice shopping, and IoT payments.

10.2.1. P2P payments:

One of the earliest applications of speech in banking was sending personal payments to contacts on a list. Due to time savings, ease of use, and the fact that they can be performed with restrained hands, most consumers choose voice commands. Players in the fintech space (PayPal, Venmo, Cash App, etc.) provide a less expensive, more streamlined method of “sending money back and forth”.

Users are drawn to digital banks (Revolut, Monzo, and Chime) by their low-to-no-fee SEPA/ACH transfers and one-tap bill-splitting capabilities. By providing a seamless voice-activated P2P payment experience, banks can attempt to retain some lost transactions by appealing to their current clientele. Otherwise, they risk falling victim to rival AI systems like Google Assistant, Apple Siri, and Amazon Alexa. In the banking industry, Royal Bank of Canada, Barclays, and Santander were among the first to offer speech recognition payments using Siri.

Customers of Garanti Bank and Westpac Banking Corporation can utilize the app and conduct various financial activities by using voice commands by integrating Siri, Alexa, Google Assistant, and other voice-enabled home devices into their banking ecosystems. BBVA's recently launched MIA (Mobile Interactive Assistant) goes further and depends on NLP (natural language processing) to transmit payments, settle invoices, or perform other tasks.

10.2.2 Voice shopping:

Consumers want contactless commerce in the post-pandemic world. Players in the tech industry are fast to respond to this new (and persistent) trend. For instance, Google and Carrefour recently agreed to collaborate on launching a voice-activated grocery shopping service in France and 30 other countries where the food retailer does business.

Customers of Carrefour will be able to connect a Google assistant to their account and then simply say the products (coffee, Coca-Cola, green apples) they want to add to the cart. The voice-based checkout and payment process will also be used. Since ardent drivers prefer talking over tapping while they commute, in-car payments and commerce are other prospective areas of interest for financial players. Driving-related voice assistant connections made up 53.3% of all in-vehicle purchases in the US last year, totalling \$212 billion across drive-thru orders, gas, supermarket shopping, and in-car entertainment. To improve the voice payment experience for their customers, banks and other financial service providers should work with OEMs.

10.2.3 IoT payments:

Last year, stores without cashiers or contactless payments seemed more like a passing trend than a practical innovation. However, when COVID-19 hit, several shops evaluated the

viability of utilizing IoT technologies to develop a fully unattended shopping experience. That new experience can include voice-enabled payments, or they can be used as an additional way of transaction authentication.

Other possible use cases for speech in the IoT payment arena include connected parking, smart vending machines, self-serve hotel rooms, smart pumps, and smart vending machines. The voice may soon become the primary payment enabler and connection inside the M2M interactivity field as the manufacturing space moves toward serviced offers and pay-as-you-use service models.

10.3 Product Viability

The voice-based payment method has the potential to transform the usual two-way relationship between customers and enterprises. It is hands-free, allows faster checkout, supports regional dialects, and is fully functional on both smart and feature phones. In the current situation, customers manually visit the bank to obtain banking services, including money transfers which are more expensive and time-consuming, rather than realizing these processes using phones/mobile.

Besides the cost savings, the proposed system also increases accessibility for people, such as the illiterate or blind, as it is based on voice, making it easier to use. The usage of voice-based payment calls ensures time efficiency. On the other hand, this scheme provides a high degree of security (in terms of authentication) since it relies on two-factor authentication steps because of the system's fast post-processing ability and security. Another strength of the scheme is its aptitude for further developments in banking services.

It is foreseen that the proposed mechanism enables various opportunities for further applications and developments via customized integrations. Voice-activated payments are far more accessible than typing out long card numbers, account numbers, IBAN numbers, and so on. A transaction can be completed more quickly and easily using voice commands. Plus, many more users can be reached. Hence, the system will be more scalable, and e-banking services will be easier to initiate. This opportunity also fetches private sector companies and non-profit organizations to implement voice-based payments.

The government and most financial institutions are trying to make online payments safer by minimizing the risks involved. The proposed software product can offer potential solutions to make online transactions safer by preventing e-frauds. Also, due to the introduction of several mobiles and online banking checks, the user experience might impact a little. Still, the proposed solution is quite viable for existing payment banks to reduce the potential risks evolved through tech-savvy fraudsters.

10.4 Market Development

The market for voice-based payments may experience significant expansion due to government initiatives taken by many nations to promote digital payment solutions and the digitalization of payment methods. Due to the preference for contactless payment methods to

prevent coronavirus transmission through touch, the COVID-19 pandemic significantly created fast development potential for the voice-based payments business. The market has grown due to several companies and stores' adoption of voice-based payment methods.

The public can be aware of the developed product through advertising campaigns and promotional activities, which may accelerate the market expansion for voice-based payments. Collaborations and strategic planning are essential for market expansion. The market for voice payments is fairly consolidated globally, with a small number of large and medium-sized companies accounting for the bulk of market revenue. Major companies such as Google Assistant, Alexa, and Siri are deploying various strategies, entering into mergers and acquisitions, strategic agreements and contracts, utilizing more cutting-edge voice payment technologies, creating, testing, and launching fresh, more effective voice payment solutions on the market.

11. Target beneficiaries

1. Voice banking gives a feasible solution to the common people, mainly in rural regions of India.
2. This application allows illiterate people to get services like transferring money and balancing inquiries using their voices to communicate with the system.
3. Voice-based payment makes banking more accessible to blind or visually impaired people.
4. Voice-based payment methods will help banks and financial institutions to increase their reach, improve customer engagement, and improve their services.

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12. Gantt Chart of activities giving clear milestones for every quarter

Quarter-wise outputs/deliverables			
Goal 1: Planning & project set-up			
Activities	Timeline (months)	Expected Outcome	Comments
- Requirement Analysis - Hiring Process - Define Role and Responsibility - Literature survey on Digital payments	1-2	Project development initiated	
Goal 2: Data collection, Model Training & debugging			
Activities	Timeline (months)	Expected Outcome	Comments

• Data collection of voices in local/rural areas in different languages. • Development of a Deep learning-based model to create a Voice assistant. • Development of an Audio-warning system in regional languages	3-12	Deep learning model in at least one local language	Initially, we can try to build a model in 2 or 3 languages. Still, it will be continued data collection and creation of a model in different languages for the project's duration.
Goal 3: Android application development			
Activities	Timeline (months)	Expected Outcome	Comments
• Training and debugging of Deep Learning Model • Core idea & UX/UI designing process of android application. • Development of a QR-based authentication system	8-12	Completion of UX/UI designing of the app	
Goal 4: Deploying & Testing			
Activities	Timeline (months)	Expected Outcome	Comments
• Model parameters, • Model configuration, • Feature pipeline, • Training dataset, • Validation dataset. • Validation of Audio-warning system.	13-16	Completion of model deployment and testing	
Goal 5: Voice Authentication & Creation of API			
Activities	Timeline (months)	Expected Outcome	Comments
• Creation of Database • Generating Unique Voice ID • Creation of API • Validation of QR-based system.	17-18	Work done related to Database management and creation of API.	
Goal 6: ALFA Testing of Application			

Activities	Timeline (months)	Expected Outcome	Comments
- Requirement review - Test deployment - Test execution - Logging defect - Bug Fixing	19-20	ALFA testing done	
Goal 7: Beta Testing			
Activities	Timeline (months)	Expected Outcome	Comments
It is performed by “end users” of the software application in a “real environment” and can be considered a form of external User Acceptance Testing. It is the final test before shipping a product to the customers. Direct feedback from customers is a significant advantage of Beta Testing. This testing helps to test the product and the customer's environment.	21-22	The app will be open for real users for testing purposes.	
Goal 8: Updating of application in more local languages			
Activities	Timeline (months)	Expected Outcome	Comments
- Data collection, Training & debugging - Deploying & Testing - Updating the Mobile application	23-24	The app will be available in different languages.	In this phase, we will develop the app in at least 2 to 3 languages, but still, we will continue the data collection and Model creation in different languages. We will repeat the project development life cycle and update the mobile application when the model is completed.

13. Please mention the deliverables under the following heads with relevant details:

- Plans to translate the project into Startups and Spin-off companies

The project's outcome would be a software solution that needs to be integrated with any existing payment app. Payment security has been a major cause of concern for RBI, and it has been instructing banks and payment apps from time to time to strengthen their payment procedures. For the same, banks are spending crores on research and development. We propose a viable and effective solution to various security threats to online banking on both browsers and mobile platforms. The idea can be easily translated into the form of an R&D-based startup providing security solutions to various online payment platforms/FIs such as multiple public/private Banks, the BHIM app, GooglePay, Paytm, etc.

- Target number of Publications, Technologies, products, IPRs, etc.

The project's outcome would be novel software solutions for online payment banks/apps. This can be patented in the form of IPRs.

- Target number of job creations

This project will generate four direct man powers, two in form JRFs and two in form research associates during the execution of the project. However, the proposal can be translated into a startup, generating several other job opportunities.

14. Budget

Head	1 st Year	2 nd Year	3 rd Year	Amount contributed by industry	Amount payable by IBITF
Equipment	9,50,000	NIL	NIL	NIL	9,50,000
Travel	1,75,000	1,75,000	1,00,000	NIL	4,50,000
Manpower	16,56,000	16,56,000	16,80,000	NIL	49,92,000
Contingencies	50,000	50,000	50,000	NIL	1,50,000
Overhead	1,50,480	1,50,480	1,46,400	NIL	4,47,360
Grand Total	29,81,480	20,31,480	19,76,400	NIL	69,89,360

A. RECURRING BUDGET

S. No	Item	Budget (Rs.)			
		<i>1st Year</i>	<i>2nd Year</i>	<i>3rd Year</i>	Total
1	Manpower (JRF, no: 03)	11,16,000	11,16,000	12,60,000	34,92,000
2	Manpower (Project Assistant, no: 01)	2,40,000	2,40,000	1,20,000 (6 months)	6,00,000
3	Manpower (Project Associate-I)	3,00,000	3,00,000	3,00,000	9,00,000
4	Travel	1,75,000	1,75,000	1,00,000	4,50,000
5	Contingencies	50,000	50,000	50,000	1,50,000
6	Institutional Overheads (8%)	1,50,480	1,50,480	1,46,400	4,47,360
RECURRING TOTAL (A)		20,31,480	20,31,480	19,76,400	60,39,360

Detailed justification for Recurring Budget heads:

1. Manpower

Roles and Responsibilities of JRF & SRF:

1. To perform an extensive literature survey related to the problem statement
2. To implement the various state-of-the-art ML/DL models
3. To develop the proposed models under the guidance of PI & Co-PI
4. To validate the proposed models for a real-time data set
5. To deploy the proposed model in real-time for end-users
6. To publish the work in reputed International Journals/Conferences and Patents.

Roles and Responsibilities of Project Assistants:

7. To collect a massive number of data sets from different parts of rural regions of India.
8. To create a voice-based specific data repository for various regional languages.
9. To create the mobile application for the proposed AI model.
10. To help the PI in project-related work

2. Travel:

1. Data Collection: Collect a huge number of real-time data sets in different regional languages from various parts of rural India for better model creation.
2. Present the research work to the PRSG committee and Conference/Symposium/Workshop in India.
3. Contingency: It will be used to register papers at International Conferences, attend Workshops/FDPs, etc. And also involve the purchase of stationery and other items required during project work.
4. Overhead: As per institute norms, the overhead charge for running project work in this institute is (8-15) %. Considering the 8% overhead charges of the annual recurring budget of the project.

B. NON-RECURRING: BUDGET FOR PERMANENT EQUIPMENT

S. No.	Equipment: Qty	Budget (Rs.)		
		1 st Year	2 nd Year	Total
1	Workstations: 04	8,00,000	NIL	8,00,000
2	Smart Phones: 05	1,00,000	NIL	1,00,000
3	Printer & Scanner: 01	50,000	NIL	50,000
NON-RECURRING TOTAL (B)		9,50,000	NIL	9,50,000

Detailed justification of Permanent Equipment:

Equipment's	Justification	Quantity	Cost of each quantity	Cost
Workstations	Dedicated workstations are required to carry out the research work by the JRFs/PIs in the field for ML/DL applications	4	2,00,000	,00,000
Smart Phone	Smartphones are used for testing and training the proposed model.	5	20,000	1,00,000
Printer & Scanner	For research-related documents printing and scanning	1	50,000	50,000

15. The expertise of PI

- a) Bio-Data of the PI and Co-PI (as per annexure 4): Attached
- b) Infrastructural Facilities available with the investigator: GPU Server is available
- c) Equipment available with the Institute/ Group/ Department/ for the project:(Related to this project): GPU Server is available for complex data computation/processing.
- d) Prior Knowledge/Publications in the relevant area: PI has three ongoing funded projects in the area of Artificial Intelligence. PI & Co-PI have various publications in reputed International Journals/Conferences in Machine Learning, Deep Learning, and Speech Processing.

Roles and Responsibilities of PI's and Co-PIs.

Name	Role	Responsibility
Dr. Nenavath Srinivas Naik	PI	(1) To develop an end-to-end voice-enabled mobile application for day-to-day digital transactions. (2) Deployment and testing of complete payment setup.
Dr. Anurag Singh	PI	(1) NLP-based system design for text-to-speech in regional languages.

		(2) Integration of NLP-based system with digital payment app
Dr. Abhishek Sharma	Co-PI	(1) Transaction data collection and model design for OTP fraud prevention. (2) Deployment of a machine learning algorithm for OTP fraud prevention.
Dr. Shrivishal Tripathy	Co-PI	(1) Transaction data collection and model design for OTP fraud prevention. (2) Deployment of a machine learning algorithm for OTP fraud prevention.
Dr. K G Srinivasa	Co-PI	(1) Development and deployment of voice and graph-based User Interfaces for digital transactions. (2) Integrating the fraud prevention system with the payment app
Dr. Santosh Kumar	Co-PI	(1) Development and deployment of voice and graph-based User Interfaces for digital transactions. (2) Integrating the fraud prevention system with the payment app
Dr. Amit Kumar Agrawal	Co-PI	(1) Customer Discovery and Profiling (2) User Validation (3) Managing Finance (4) Developing Marketing and Promotional plan